Engineering Applications of Artificial Intelligence 122 (2023) 106087

Contents lists available at ScienceDirect

Engineering Applications of Artificial Intelligence

journal homepage: www.elsevier.com/locate/engappai

Survey Paper

A systematic survey on explainable AI applied to fake news detection

Athira A.B. *, S.D. Madhu Kumar, Anu Mary Chacko

Department of Computer Science and Engineering, National Institute of Technology Calicut, kozhikode, 680541, Kerala, India

ARTICLE INFO

Keywords:
Fake news detection
XAI
Machine learning
Explainable fake news detection
Fake news identification

ABSTRACT

The exponential proliferation of fake news in recent years has emphasized the demand for automated fake news detection. Several techniques for detecting fake news have yielded encouraging results. However, these detection systems lack explainability i.e., providing the reason for their prediction. The critical advantage of explainability is the identification of bias and discrimination in detection algorithms. There are very few surveys conducted on the area of explainable AI applied to fake news detection. All of theses surveys summarize the existing methods in this area. Most of them are limited to the discussion of specific topics like datasets, evaluation methods, and potential future applications. In contrast, this survey looks at existing explainable AI methods and highlights the current state of the art in explainable fake news detection. We identify and enumerate a few open research problems based on our review of the existing explainable fake news detection techniques. We group the existing work in this area, by viewing it from four different perspectives: features used for the classification, explanation type, explainee type, and the metric used for explainability evaluation. The potential research topics in the above four groups which are unexplored so far and which need attention are also listed in this paper.

1. Introduction

Online social media platforms are becoming increasingly popular across different generations, and these are eroding the traditionally authoritative voices of the news industry. In social media, people discuss and express their opinions about the news by leaving comments. Recently much of the news material on social media are found intentionally misleading. The phrase "fake news" is used to describe such malicious news items. The 2016 presidential election in the United States is said to have been impacted by fake news in social media. Following election, this phrase has become a common term which is referred on a daily basis everywhere. For many reasons, fake news identification is a technological challenge. News Contents are easily generated and spread through social media platforms, resulting in a large amount of data to evaluate. The vastness of online material, which covers many topics, adds to the task's difficulty. It has prompted the researchers to work towards automatic fake news detection. For detecting fake news, many researchers have suggested a range of machine learning and deep learning approaches. Beside their promising results, the models lack explainability (reason for the prediction made). In this paper, we provide a comprehensive overview of the research on explainable fake news detection to date. Finally, we recommend several potential research areas for assessing fake news on social media.

1.1. Defining fake news

The most challenging aspect of detecting fake news is defining fake news. There is no standard definition for fake news. According to Wikipedia, "Fake news is defined as false or misleading information presented as news, Its goal is frequently to harm a person's or entity's reputation or to profit from ad revenue" (Wikipedia, 2021). Zhou and Zafarani (2020) differentiated fake and fake-news-related phrases based on authenticity, purpose, and whether or not it was news. The authors gave two different definitions (broad and narrow definitions) for fake news. "Fake news is false news", according to the broad definition, Zhou and Zafarani (2020), and "fake news is intentionally false news published by a news outlet" (Zhou and Zafarani, 2020) according to the narrow definition.

In Zhang and Ghorbani (2020) the authors have divided the current fake news identification methods into two: *practical-based approaches and research-based approaches*. The division is done according to the audience of the fake news. While practical-based approaches are helpful for naive users, researchers or academicians follow research-based approaches.

Use of fact-checking websites is the most prevalent method of applying the practical-based approach. Table 1 lists the various fact-checking websites that are currently in use and their detection technologies. Human resources-based methods are involved in the majority of fact-checking websites. However, human fact-checking is limited by time,

* Corresponding author.
E-mail address: athira.bala.sree@gmail.com (Athira A.B.).

https://doi.org/10.1016/j.engappai.2023.106087
Received 20 October 2022; Received in revised form 5 February 2023; Accepted 2 March 2023
Available online 20 March 2023

Table 1
List of fact-checking websites that are currently in use.

Fact-checking websites	Detection methods
Media BiasFactCheck (MBFC News) FactCheck.org (Media Bias, 2021)	Manual detection
TruthOrFiction.com (2021)	Manual detection
Lead Stories (2021)	Trendolizer technology and Manual detection
FullFact.org (2021)	Detection by fact checkers
Classify.news (2021)	Machine learning algorithms
Factmata.com (2021)	Artificial Intelligence
Politifact.com (2021)	Manual detection
Snopes.com (2021)	Manual detection
Hoax-Slayer.com (2021)	Manual detection

as a large amount of data is created daily and spread quickly. It also faces scalability issues. Thus, the difficulties imposed by manual fact-checking approaches paved the way towards the research based approaches or automatic detection of fake news.

1.2. Automatic fake news detection

The majority of published research works in this area, treat fake news detection as a binary classification problem. As a result, a general definition for detecting fake news is formulated as stated below.

Given the features extracted from the news item x, the fake news detection model aims to classify the news item as real or fake. F denotes the fake news detection model. Let a_i represents a feature extracted from the news item x. where i = 1,2,3, … .

$$F(a_i) = \begin{cases} 0 & \text{if x is a fake news} \\ 1 & \text{otherwise} \end{cases} \tag{1}$$

A number of surveys on the detection of fake news are available in the current literature. In the survey (Shu et al., 2019a), authors presented a review of the identification of fake news from a data mining perspective. The authors have highlighted detection of fake news from four different perspectives: *incorrect knowledge in the news, the writing style used in the news, the news's propagation pattern, and the credibility of the news source*. The authors discuss the negative impact of fake news and current detection algorithms in Zhang and Ghorbani (2020). The authors of Mohseni et al. (2020) presented a summary of published research articles in the field of fake news detection from 2016 to 2020.

Based on the literature review we find that, another way to classify the problem of identifying fake news is to categorize it as single-modal or multi-modal fake news detection, depending on how many features are involved.

1.3. Single-modal fake news detection

The features used for fake news detection are divided into three categories: The three categories are *user features, news content features, and social context features*. In single-modal fake news detection, a single type of feature is analyzed. Fig. 1 gives an overview of the various features used in single-modal fake news detection. User features include information in the user profiles, credibility-related information, and user behaviour patterns. The features of news content are derived from the analysis of words, sentences, and news content. These are then referred to as linguistic-based and syntactic-based features. There is another class of news content features with focus on style-based and visual-based features. The last category, social context features, is derived from the propagation pattern of the news and the interactions of social media users. It includes temporal, distribution, and network-based features.

We observed that news content features are used in the most of the single-modal methods. Sticking only to a single kind of feature will not ensure accuracy. When the writing style is modified, these approaches become irrelevant. The addition of auxiliary information (combining different features) aids in the resolution of this approach, and thus has resulted in the a multi-modal fake news detection system.

1.4. Multi-modal fake news detection

In the case of multi-modal fake news detection, various feature combinations are examined. Recent research works developed in multi-modal detection combines two kinds of feature sets. (1) a mix of visual and textual elements, and (2) user profile and news content elements.

1.4.1. Detection by combining textual and visual features

In the literature, many research works have used a combination of textual and visual features for detecting fake news. A brief overview of the feature extraction and classification technology used in these approaches is presented here.

A model or microblog news verification that incorporates both visual and textual information was proposed in Jin et al. (2017). The authors proposed two kinds of image features. (1) statistical features and (2) visual features from the news. The statistical features capture the image distribution patterns. The visual characteristics disclose hidden distributions in news events. They have used seven statistical features and five visual features. Fig. 2 show the details of the image features used in the model. Traditional classifiers such as support vector machine (SVM), logistic regression, K star, and random forest are used for classification. A real-world multimedia data set created using news events in Sina Weibo (a micro-blogging site also referred as "Chinese Twitter") is used for the performance evaluation.

Research papers Singhal et al. (2019, 2020), and Shah and Kobti (2020) investigate multimodal fake news detection using deep learning models. The authors of Singhal et al. (2019) presented *SpotFake*, a multimodal framework for detecting fake news that makes use of both the textual and visual features of an article. For learning textual characteristics, they employed Bidirectional Encoder Representations from Transformers (BERT). VGG-19 is used to learn image characteristics (pre-trained on ImageNet). The studies were carried out by using two publicly accessible datasets, Twitter and Weibo.

SpotFake+ (Singhal et al., 2020), is an enhanced version of *SpotFake* (Singhal et al., 2019). The first version of the model does not take into account the context information in the text. They were unable to capture the elements of the image modality that may have sought to emphasize particular facts. *SpotFake+* used transfer learning to capture the textual and visual components of a news article. The feature extraction is done using the XLNet and VGG-19. After that, the features are combined and used for classification. The data set used is FakeNewsNet (FakeNewsNet, 2021).

The model proposed in Shah and Kobti (2020) was separated into three primary components by the authors. The first is *a sentiment analysis-based textual feature extractor*, which uses sentiment analysis to extract meaningful content from textual data. The second component is the *visual feature extractor*, which extracts visual features from the post using preprocessing techniques and a segmentation algorithm. A cultural algorithm was used to extract the optimum features from the textual and visual feature representations. The last component is *a fake news detector*, which classifies the news into fake or real. Through the implementation of Cultural algorithm for optimal feature extraction, the authors have outperformed related approaches.

In Tanwar and Sharma (2020) authors extracted visual characteristics using three common CNN architectures: VGG-19, ResNet50, and InceptionV3. Bidirectional LSTM was used to extract textual characteristics. The combination of textual and visual features was then utilized to classify fake news

In Singh et al. (2021), the authors used three modalities: textual, visual, and a combination of text and visual features for fake news detection. Predictive analysis on both textual and visual features is carried out to determine their association with fake news. Many traditional machine learning algorithms were applied to classify the news items.

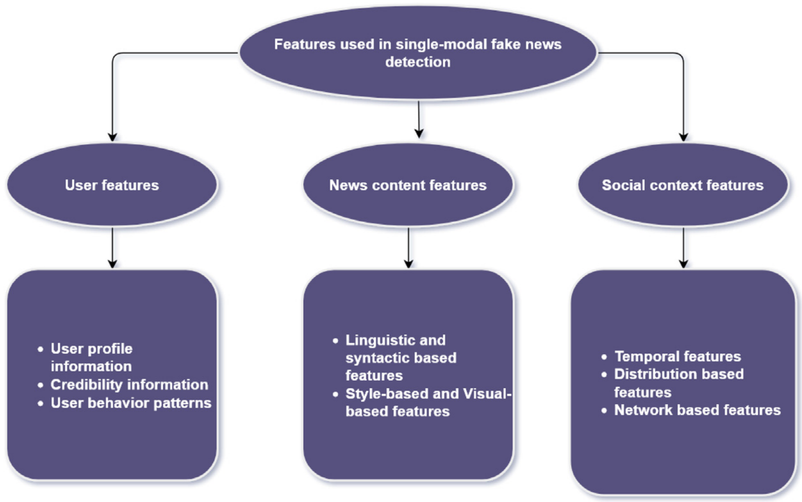

Fig. 1. Categories of features used in single-modal fake news detection.

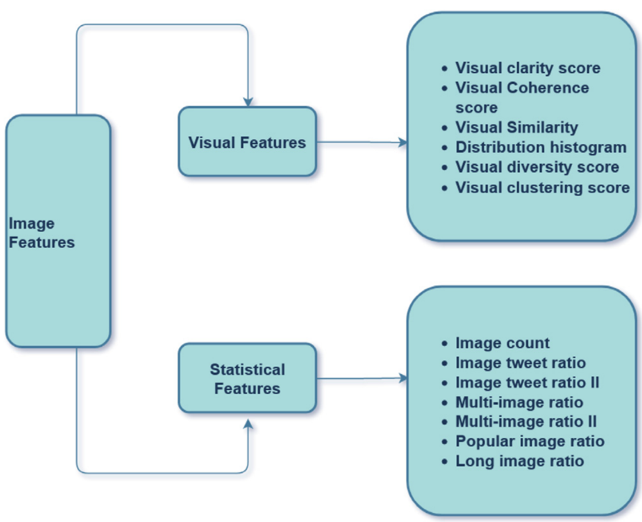

Fig. 2. Overview of image features used in Jin et al. (2017).

1.4.2. Detection based on user profile characteristics

Methods for detecting fake news based on user features are discussed in this section. User interaction is critical in detecting fake news. There is a subset of users who readily believe any fake news and have features that differ from those who believe real news (Shu et al., 2018). The authors of Shu et al. (2019c) extracted implicit and explicit user profile features from the user profiles of those who are inclined to spread fake news. They proved the use of these derived features in the categorization of fake news in their study. Their results surpassed prior algorithms that employed news content to classify fake news, which prompted the researchers to develop a detection method that used user attributes and news content aspects.

The authors of Ruchansky et al. (2017) introduced a new model *Capture, Score, and Integrate (CSI)* that aims to capture three characteristics of fake news. CSI is composed of three modules: Capture, Score, and Integrate. In the first module, Capture temporal patterns of the user's activity extracted along with the text. The second module, Score, assigns a score to each user. Finally, in the Integrate module response, text and source information from the previous two modules are combined and classification is performed. A similar approach was proposed by Liu and Wu (2020) for Fake News Early Detection named, *FNED*. In this case, the authors retrieved textual and user characteristics from the users' responses, as well as their accompanying user profiles. A status-sensitive crowd response feature extractor is implemented for this purpose. They have used CNN-based news classifiers for labelling the user responses. In order to enhance the performance of the model, they employed a PU Learning framework (learning from positive and unlabelled examples).

Discussion

Regardless of the modality of the features used for training, all the present research is pursuing AI-based solutions since deep learning (DL) systems have demonstrated unquestionable performance compared to traditional machine learning techniques. However, the black-box nature of the methods makes it hard for a naive user to grasp the basis for prediction. When comparing fake news detection approaches using deep learning models with the standard machine learning models, their performance is hampered by their lack of explainability. So to avoid bias and improve performance, the detection model must have transparency in its decision-making process. Thus, explainable AI (XAI) approaches help to detect fake news since they enable us to comprehend better and explain the model's prediction.

At present, the research on explainable fake news detection models is very limited. The existing surveys conducted in this area summarize the state-of-the-art models. These reviews discuss mainly the datasets used, explainability models used, and potential applications in the future. In comparison with existing surveys, our survey begins by examining the explainable AI techniques that are now in use, which is crucial for the reader to become familiar with the methods used in XAI. Concepts related to XAI are covered in Section 2. In Section 3 we list the existing models for explainable fake news detection. We have also classified the present research works in this field by assessing them from four different perspectives. A list of prospective applications for the future are also discussed in this review.

2. Explainable AI systems (XAI)

The AI system is deemed an XAI system, if it can explain decisions taken. In this section we describe the most prevalent XAI concepts.

2.1. Interpretability

2.1.1. Intrinsic interpretability

Intrinsically interpretable machine learning methods are interpretable due to their simple structure. Self-explanatory models

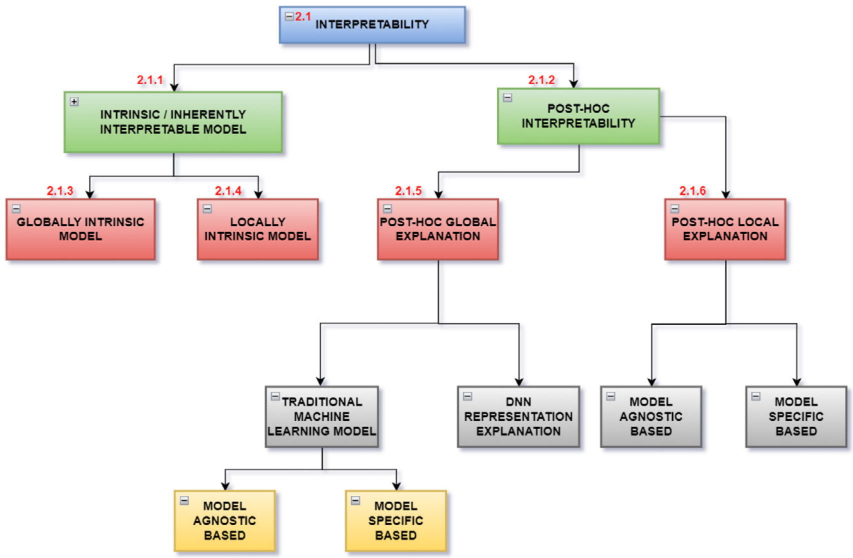

Fig. 3. A summary of the various interpretability approaches.

such as decision trees, linear models, attention-based models, and rule-based models are examples of the intrinsically interpretable model. Even though these models provide precise explanations, they forfeit the prediction performance.

2.1.2. Post-hoc interpretability

In the post-hoc interpretable model, interpretation methods are deployed after model training. Therefore, to explain the preceding model, a second model has to be built. The models are constrained by their approximate nature, but the models' accuracy is preserved. The authors of Singhal et al. (2019) further divided interpretability into global and local interpretability. Local interpretability is about understanding a single prediction of a model. Global interpretability is concerned with comprehending how the model makes decisions based on a complete overview of its characteristics and learned components, such as weights, other parameters, and structures.

The different interpretability approaches listed in Fig. 3. are explained next.

2.1.3. Intrinsic global interpretability

Globally intrinsic models have two approaches:

1. Adding constraints: Examples for addition of constraints include pruning of the decision trees, restricting the number of features used for predictions, and using features with monotonic relations.
2. Interpretable model extraction/mimic learning: A complicated model is simulated using simple models such as decision trees, rule based models, and linear models.

2.1.4. Intrinsic local interpretability

Attention mechanisms are used in the locally intrinsic model to give users an understandable justification for a specific prediction.

2.1.5. Post-hoc global explanation

Patterns are learned from the training data and retrained into model structures and parameters. There are two kinds of approaches. These are Traditional machine learning explanations and Deep Neural Network (DNN) representation explanations.

1. Traditional machine learning explanations: Feature importance is utilized, which assesses each feature's contribution to the prediction.
    - Model-agnostic explanation:- The model-agnostic explanation is a method that separates the explanations from the trained model. An example is Permutation Feature Importance: How well the model's prediction accuracy deviates after permuting the feature values determines the significance of a given feature to the model's performance.
    - Model-specific explanation:- In the model-specific explanation, the internal structure and parameters are examined to derive the explanation. A generalized linear model (GIM) and tree-based ensemble models are examples. GIM includes linear regression and logistic regression. Gradient boosting machines, random forest, and XGBoost (2021) are examples of tree ensemble models.
2. Deep Neural Network (DNN) representation explanation:
    - Explanation of Convolution Neural Network (CNN) representation: It involves finding preferred input for neurons in a specific area.
    - Explanation of Recurrent Neural Network (RNN) representation: Language modelling that predicts the next token given the previous token is used to examine the representation learned by the RNN.

2.1.6. Post-hoc local explanation

1. Model-agnostic explanation: Explanations are generated without accessing the internal parameters of the model.
    - Local approximation based explanation: Any black-box model is approximated with a local, white-box model to explain each individual prediction. The white box model's parameters were then used to get the feature contribution score.
    - Perturbation based explanation: The feature's contribution is determined by looking at how the prediction score changes when the feature is changed. Counterfactual explanations are the outcomes. It can be accomplished by either omission or occlusion. Omission removes a feature from the input, whereas occlusion replaces the feature with a reference value. In case of word embedding, zero is used as reference whereas in image grey value used.
2. Model-specific explanation: In a model-specific explanation, explanations are created that are specific to the model.

- Back-propagation based method: The gradient is determined for a given output with respect to the input utilizing back-propagation to determine a feature's contribution.
- Mask perturbation: The model-agnostic-based perturbation is computationally expensive. DNN-specific perturbations were obtained using Mask perturbation and a gradient descent optimization framework.
- Investigation of deep representation: The DNN's deep representation offers a wealth of information for interpretation. It lowers the likelihood of producing artifacts and produces more meaningful explanations. The deep representation is ignored in a perturbation or back-propagation-based approach.

We find that the explanations detailed above are geared for developers or researchers as their target audience. If the explainee is a naive user, this sort of explanations are not appreciated much. The authors of Du et al. (2019) provided some user-friendly explanations. It includes *contrastive explanations, counter-factual explanations, selective explanations, credible explanations, and conversational explanations*.

1. Contrastive explanations: Differential explanation is a term used to describe this type of explanation. Instead of providing a reason why a particular prediction was made, the authors provide an explanation on why one prediction was made over the other. "Why Q rather than R?" Q is the prediction that needs explanation, and R is the other case (Du et al., 2019). This is also called Counter-factual explanations.
2. Selective explanations: The user is given the top factors that influenced the prediction.
3. Credible explanations: Good explanations must be consistent with the consumers' existing understanding. However, while these explanations may be accurate, the machine learning model does not use correct evidence when making decisions.
4. Conversational explanations: In the case of conversational explanations, an explanation is presented as a discussion between the explainer and the explainee.

2.2. A comparison of explainability and interpretability

According to the authors of Atakishiyev et al. (2020) Interpretability and explainability are two different concepts. Based on Atakishiyev et al. (2020), a human user can inspect an interpretable model, whereas explainability has to reason for prediction. The authors grouped research trends in XAI into four: *concurrently constructed explanations, post-hoc explanations (which can be model-dependent or model independent), application-dependent vs. generic explanation, and classification based on the levels of the explanation.*

1. Concurrently constructed explanations: Along with their fundamental purpose, models attempt to generate explanations. An example technique is learning attention weight from the input.
2. Post-hoc explanations: The core concept is to approximate a trained model and generate explanations. There are two approaches with post-hoc explanations: (1) model-dependent and (2) model-independent. In a model-dependent method, we have access to the model's internals. Model-independent techniques do not have access to the model's internals. Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017) and Local Interpretable Model-Agnostic Explanations (LIME) (Ribeiro et al., 2016) are two model-dependent and model-independent techniques, respectively.
3. Application-dependent vs. Generic explanation Explanations: are created based on the type of explainee. In an application-dependent approach, the explainee is implicitly assumed to know about the application, hence the domain vocabulary is used.
4. Classification based on the levels of the explanation, categorized the types of explanations into four levels.
   Level 1: This category includes models that only employ one sort of explanation. The majority of models provide a post-hoc explanation.
   Level 2: An alternative explanation is offered in addition to the single kind of explanation. If one explanation is not clear, we can deduce the classification reason from the alternative explanation. For example, let us say a model wants to classify an animal image as a dog or cat. A heat-map can be used to describe why a model classified the animal image as a dog. Along with the heat-map, a textual explanation is provided as an alternative explanation.
   Level 3: The explainee's understanding is taken into account while offering an explanation. At the moment, none of the models fit into this category.
   Level 4: This category includes models that allow for interaction between the model and the explainee.

2.3. Discussion on explainability approaches

The authors of Belle and Papantonis (2021) have done a generalized discussion on explainable machine learning, which includes ideas of transparency, methods on assessing explainability, and the kinds of explanation (post-hoc explanations). The authors discuss different types of post-hoc explanations that cover text, visual, local explanations, explanations using example, explanations using simplification, and feature relevance explanations. They proposed five frameworks that characterize advancements in explainable machine learning. These are *(1) taxonomic framework, (2) Transparency framework, (3) Explainable machine learning capability framework, (4) Explanation type framework and (5) Data scientist strategy framework (Belle and Papantonis, 2021)*. The authors of Belle and Papantonis (2021) have listed a detailed overview of the explainability approaches for deep learning models.

2.4. Evaluating explainable systems

A detailed study and an approach for evaluating explainable machine learning systems are discussed in Mohseni et al. (2021). The authors grouped current design and assessment techniques that arrange literature into three categories: *(1) design goals, (2) evaluation methodologies, and (3) the XAI system's targeted users*. The design goal is organized with respect to three different user groups (AI end users, data experts, and AI experts). Each has its own set of design objectives. Evaluation methods used for evaluating machine learning models are *user mental models, user trust and reliance, explanation usefulness and satisfaction, human-machine task performance, and computational measurements* (Mohseni et al., 2021). An interdisciplinary team of researchers created an XAI system for fake news detection for non-expert daily newsreaders as a case study to illustrate how the framework may be used in practice.

2.5. Need for XAI in fake news detection systems

Fake news detection methodologies have attracted the attention of research community in recent years. Several approaches have been developed based on machine learning and deep learning methods for fake news detection. Although this fake news detection model has shown promising results, the explainability of such detection is still largely unknown, rendering the conclusions uncertain. Compared to earlier detection models, new approaches use additional information (social context, multimedia content, and user information) to improve detection. This additional information can provide explainability (reasoning why a piece of news is fake or not).

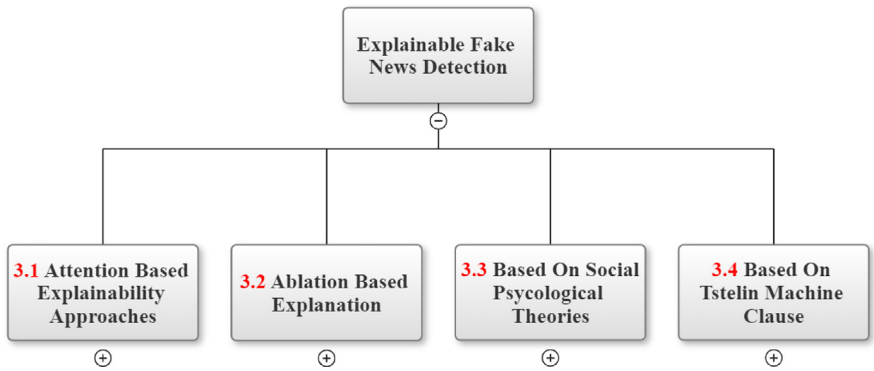

Fig. 4. Explainable fake news detection methods.

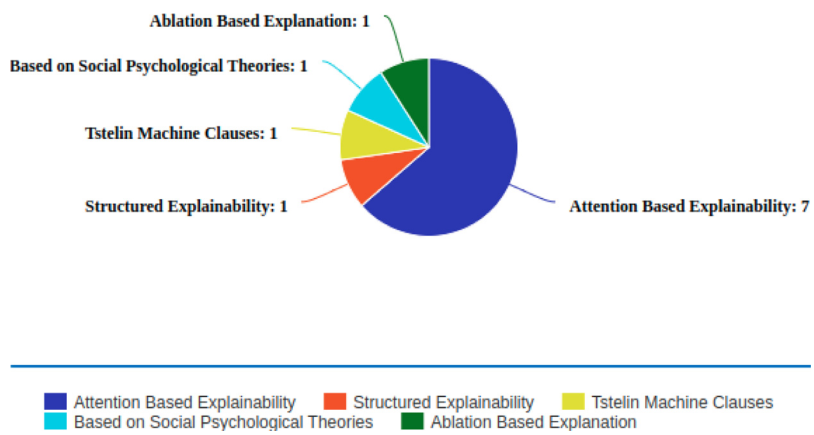

Fig. 5. An overview of statistics of explainable fake news detection models.

3. Related work: Explainable fake news detection

This Section reviews the literature and provides an overview of the various methods proposed for explainable fake news detection. We divided the related work based on their explainability approaches. The techniques and features used for each proposed model are given with all relevant details. Fig. 5 gives an overview of the statistics of the existing approaches based on explainabilty approaches used.

Among the available models in explainable fake news detection, seven of the eleven models use attention-based models to provide explainability (see Fig. 6). The other four methods are *structured explainability, social psychological theories, Tsetlin machine clause, and ablation-based explanation.* The tree diagram for the explainable fake news detection models is shown in Fig. 4.

3.1. Attention based explainability approaches

The related papers that employed attention mechanisms to explain the prediction of fake news are described below.

3.1.1. XFake (Yang et al., 2019)

In contrast to other existing approaches, *XFake* (Yang et al., 2019) could simultaneously analyse news items from a variety of perspectives. *XFake* (Yang et al., 2019) proposes three different frameworks: *MIMIC (Yang et al., 2019), ATTN (Yang et al., 2019), and PERT (Yang et al., 2019)*. Using these three frameworks, it provided a prediction score for news fakeness along with an explanation for the predictions. Table 2 presents details of the *XFake*.

Discussion: The *XFake* dataset is quite small and it includes information related to politics. The model has used only the textual content for analysis. News with multimedia content is more appealing to people

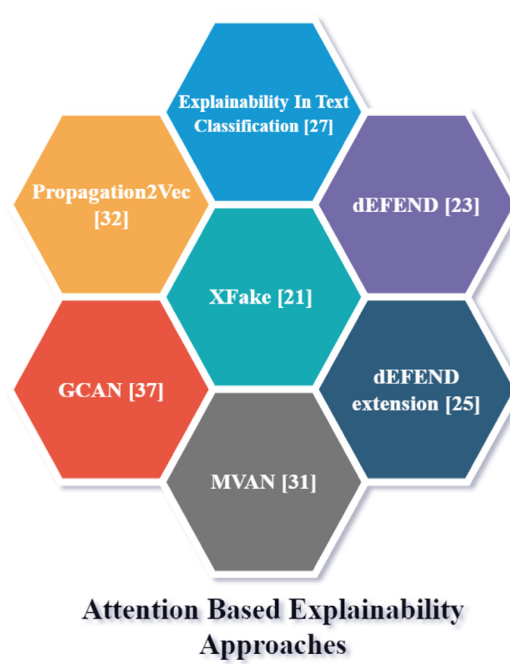

Fig. 6. Various attention based explainable fake news detection methods.

than news in plain text. This cognitive ability is being abused by those who distribute false information. Thus, expanding the model to include multimedia content analysis is necessary for increasing the potential use of this model.

Table 2
An overview of the *XFake* model.

Model used	*MIMIC* evaluates each news item through its attribute information. Tree ensemble models mimic the neural network's performance. Transparency to the model is obtained using the constructed tree and the attribute's importance scores. *ATTN* investigates statements through their semantic meaning and explanations obtained using self attention mechanisms. *PERT* studies the news statement's linguistic features for detection through its semantic meaning and provides the perturbation-based explanation. With these three self-explanatory models, relevant explanations are derived for interpreting the detection results. *XFake* has implemented visual explanation for prediction and explanation using histogram, heatmap, and interactive diagrams for ensemble trees.
Evaluation metric for explainability	Amazon Mechanical Turk (AMT) was employed to conduct human evaluations to validate the quality of explanations.
Data set	Experiments were conducted on real world data crawled from the PoltiFact
Accuracy	67.1%, 67.3%, 53.2% accuracy for *MIMIC*, *ATTN* and *PERT* respectively
Critical appraisal of the tools	The *XFake* is the first explainable fake news detection tool that looks at text from a variety of perspectives and helps provide users with a more accurate interpretation. The credibility checker of *XFake* is helpful for social media posts with more textual content. The *MIMIC* framework simultaneously keeps the excellent performance from neural networks and good explainability from tree ensemble models. The *ATTN* framework can capture global relationships between different words efficiently and weights assigned to the words. The *PERT* framework replaces the feature value with random noise drawn from the same distribution as the original one, and then the prediction difference is computed. This prediction difference is the significant score for the corresponding feature. The output of *XFake* is the news prediction as a unified score, indicating its probability of being fake. The results from three different frameworks are combined, and then weights for combination are tuned on a validation set. The higher the prediction score, the higher is the probability of fake news.

Table 3
An overview of the *dEFEND* model.

Model used	The authors employed a sentence comment co-attention sub-network to boost fake news detection performance to capture the intrinsic explainability of news phrases and user comments. The *dEFEND (Cui et al., 2019)* algorithm module provides search capabilities for searching news propagation networks, trending news, top claims, and related news. Simultaneously it shows the detection result and explanations. It displays both the detection results and the explanations at the same time.
Evaluation metric for explainability	dEFEND (Shu et al., 2019a) is compared with Hierarchical Attention Networks (HAN) (Shu et al., 2019b) (A fact-checking tool called claimbuster, 2021 is used to acquire the rank list of all check-worthy sentences in a piece of news content). They compared it to HSA-BLSTM (hierarchical neural network combined with social information) (Cui et al., 2019) to assess user feedback (Amazon Mechanical Turk recruited to evaluate the explainability rank list of the comments).
Data set	FakeNewsNet data set is used for the demonstration.
Accuracy	Accuracy Outperforms related works by at least 5.33%
Critical appraisal of the tool	The sentence comment co-attention network in the *dEFEND* model captures the semantic affinities between news sentences and user comments. The representations of user comments and news sentences that aid in identifying fake news are given high weights.

3.1.2. dEFEND (Shu et al., 2019a)

dEFEND (Explainable FakE News Detection) an explainable fake news detection system was created by Shu et al. (2019a) and Cui et al. (2019). Information obtained from user comments can be vital in determining fake news. As a result, the *dEFEND* model may be used on social media platforms where user comments determine news credibility. Table 3 presents details of the *dEFEND*.

Discussion: For the purpose of identifying fake news, the model examines the comments and the textual content. Comments in the social media posts include text, hashtags, and emojis. Emoji inclusion is equally significant because it aids in understanding the context of the comments. This model has also restricted its analysis to textual features. A credibility check of the comments corresponding to the news items is also missing.

Table 4
An overview of the Extension of dEFEND model.

Model used	Comment credibility check achieved with MultinominalNB, which classifies comments into junky or utility. Further analysis for fake news identification involves news sentences and utility comments.
Accuracy	Accuracy increased above 90% for all machine learning algorithm
Critical appraisal of the tool	The *dEFEND* model was applied to the filtered comments and news sentences after the multinomial Naive Bayes classifier filtered out the junk comments to identify fake news.

3.1.3. Extension of dEFEND (Sharma and Sharma, 2021)

The authors of research work (Sharma and Sharma, 2021) used the methodology of the *dEFEND* (Yang et al., 2019) and the functionality of comment credibility check before attempting fake news detection. Table 4 presents details of the *Extension of dEFEND*.

Discussion: Multi-modal properties are not taken into account by the model for analysis. Although the authenticity of the comments made in response to news items is taken into consideration, the credibility of the people who makes comments is not considered. Analysis of Emojis in the comments is also essential to increase the feasibility of this model.

3.1.4. Propagation2Vec (Silva et al., 2021)

The training of *Propagation2Vec (Silva et al., 2021)* uses both a partial and a complete propagation network of labelled news items. Complete propagation network constructed with the partial information available in early propagation networks. The analysis is done on features extracted which include user-based, textual, and temporal characteristics. Table 5 presents details of the *Propagation2Vec*.

Discussion The model performed the fake news classification by observing the way how fake news are propagated. For this, node level attributes like user credibility are utilized. The number of followers and verification status determine a user's credibility. We cannot blindly trust a user based only on the aforementioned factors, as there are automatic ways for generating number of followers. The other features like text content and visual features of news contents must be taken into account in order to improve the accuracy of the results of this model.

3.1.5. Explainability in text classification (Kurasinski and Mihailescu, 2020)

The authors of Kurasinski and Mihailescu (2020) proposed a model that focuses on linguistic-based modalities for fake news detection. It

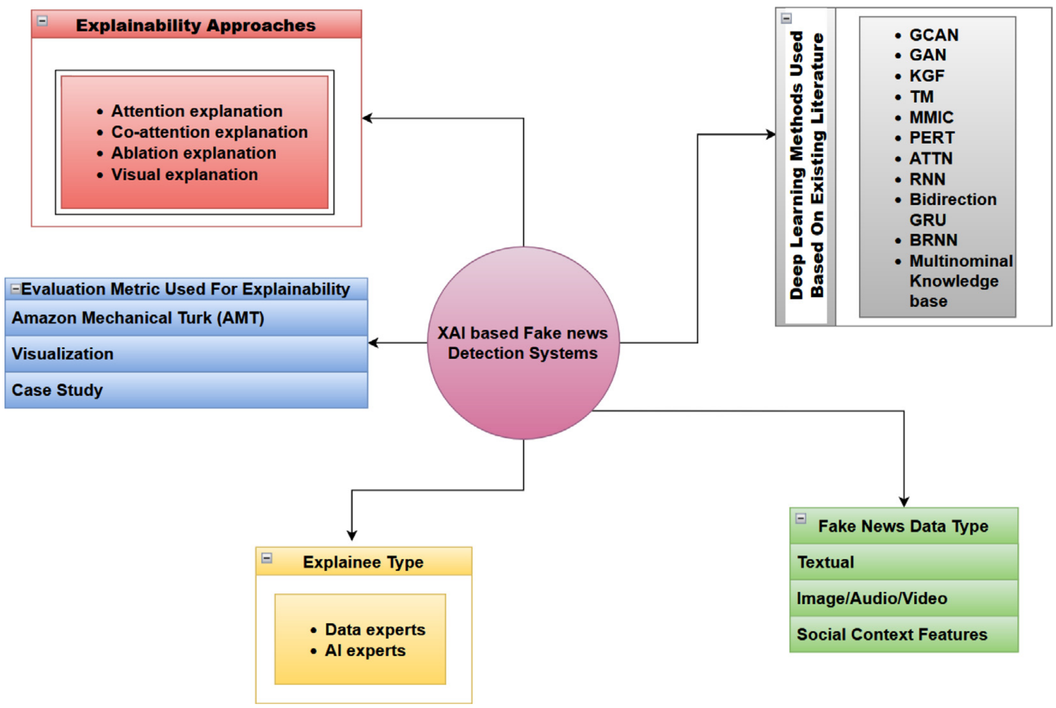

Fig. 7. Generic methodology for explainable fake news detection.

uses two deep learning models for the classification. Table 6 presents the details about the explainability in text classification.

Discussion: At the moment, the model analyses text features in news content to detect fake news. Such approaches are possible only if the news items follow a pattern. Fake news spreaders have begun to create news content in a variety of styles. As a result, other related features must be included to improve accuracy of the results.

3.1.6. MVAN (Multi-View Attention Network) (Ni et al., 2021)

MVAN utilizes source tweet and dissemination structure information for the fake news classification. Each tweet's propagation structure graph is composed of the retweeting user's information and their feature vectors. Table 7 present the details of the *MVAN* model.

Discussion The current model uses tweet text and some user profile features for classification. We can include auxiliary information like tweet comments and visual features to improve the performance.

3.1.7. GCAN (Graph-aware Co-Attention Network) (Lu and Li, 2020)

The authors of Lu and Li (2020) proposed a unique model called GCAN (Graph-aware Co-Attention Network) (Lu and Li, 2020). The algorithm predicts if the news is false or real based on the brief text contents of the tweet, user retweet sequence, and user profiles. Table 8 presents the details of the *GCAN* model.

Discussion: The short-text tweets and the corresponding sequence of retweet users without text comments are used to predict whether the source tweet is fake or not, as well as to generate explanations by emphasizing evidence from suspicious retweeters and the words they concern. The model performed classification, focusing mainly on the sequence of retweet users. Everyone has different interests, so they may retweet news that are of interest to them. To prevent biased outcomes, it is also essential to analyse news content.

3.2. Ablation based explanations

3.2.1. A language-based fake news detection using interpretable features

Discussion: The classification of fake news makes use of interpretable language characteristics. This model is language-specific. The addition of multi-modal features can further improve the results.

Table 5
An overview of the *Propagation2Vec* model.

Model used	The model uses a hierarchical attention mechanism for assigning weights for nodes and cascades and provides explainability.
Evaluation metric for explainability	An ablation study was conducted, which studied the contribution of the different modules of the Propagation2Vec (Silva et al., 2021).
Data set	Experiments are performed on Fake News Net data set
Accuracy	Outperformed 89.2% for dataset Gossipcop
Critical appraisal of the tool	Most of the existing work relies on complete propagation networks to build their model, which takes more time for fake news identification. However, early detection of fake news in the health domain is essential to avoid the significant harm it can cause. *Propagation2Vec* model is helpful in such situations. The model's accuracy has improved by the computation of attention weights at the node and cascade levels. At the node level, the model has estimated attention weights for the user, text, and temporal characteristics.

Table 9 presents the details of the language-based fake news detection model.

3.2.2. Based on social psychological theories

The authors of Zhou and Zafarani (2019) presented a pattern-driven fake news detection algorithm relying on a network. Since the spread of fake news in networks differs from real news, this study analyses a homogeneous friendship network and discovers patterns based on social psychological theories. Explanation given based on social psychological theories. The classification algorithm makes use of patterns as features. Table 10 presents the details of the model.

Discussion: With this model, early fake news detection (which deals with the identification of fake news before it spreads) is not achievable. The model used patterns found in English news text for the classification. If the fake news spreader changes his style, the patterns

Table 6
An overview of the explainability in text classification.

Model used	The first is the BiDir-LSTM-CNN, which combines CNN (convolutional neural networks) with BiDir-LSTM (bidirectional recurrent neural networks), while the second is the BERT (bidirectional encoder representation from transformers). Then two models were trained with preprocessed data sets. They are represented as colour-coded texts that indicate the network's interest in each processed word after extracting attention weights. When identifying false news texts, this visualization helps understanding how various models allocate their attention during classification
Evaluation metric for explainability	Comparison done between the two models.
Data set	Demonstration done using the data set Fake News Corpus.
Accuracy	Achieved 85% of accuracy
Critical appraisal of the tool	Unlike other baseline models, attention weights are colour coded. Important words which are coloured have a significant impact on the classification tasks. The model has provided statistics on attention distribution among the words. This approach can be helpful if the end user is not a subject matter expert.

Table 7
An overview of the *MVAN* model.

Model used	The model uses two attention networks, a text semantic attention network, to obtain the semantic representation of the source tweet information. Then a propagation structure attention network captures hidden details in the propagation structure of a tweet. The prediction model combines two representations and uses them to provide detection results.
Evaluation metric for explainability	The relative importance of each module of *MVAN* (Ni et al., 2021) was verified using an ablation study. Interpretability is visualized based on the attention scores of the source text.
Data set	Experiments done on datasets Twitter 15 and Twitter 16.
Accuracy	Outperform state-of-the-art methods by 2.5% in accuracy.
Critical appraisal of the tool	*MVAN* can detect fake news early, and the text semantic attention module is helpful for other text classification tasks on social media, like sentiment classification, topic classification, and insult detection. The propagation structure attention method is applied to identify those who distribute misinformation.

Table 8
Overview of the *GCAN* model.

Model used	First, the model extracted the user's characteristics from their profiles and social activities and learned the word embeddings from the original short text. The convolutional and recurrent neural networks were utilized for learning the representation of retweet propagation based on user characteristics and created a graph to simulate user interactions. The graph-aware representation of user interactions is learned using the graph convolution network. Also developed a dual co-attention method that learned the relationship between the source tweet and retweet propagation and the co-influence between the source tweet and user interactions. Binary classification was performed using the learned embeddings. The dual co-attention mechanism produces appropriate explanations.
Data set	Experiments done on datasets Twitter 15 and Twitter 16.
Accuracy	Outperformed state-of-the-art methods by 16% in accuracy
Critical appraisal of the tool	The model's primary goal is to forecast how reliable the source tweet will be. Explanations are produced by emphasizing the evidence about suspicious retweeters and the terms they concern.

Table 9
An overview of the model discussed in Qiao et al. (2020).

Model used	In Qiao et al. (2020) authors used CoCoGen (Complexity Contour Generator), a computational method that uses a sliding window approach to compute within-text distributions of feature scores to analyze raw texts from fake news datasets. Classified data using BRNN (Bidirectional Recurrent Neural Network) classifiers with GRU (Gated Recurrent Unit). The classifiers trained on interpretable features outperformed existing techniques using Recurrent Neural Network, Convolutional Neural Network, and Capsule Networks, that employ word embeddings to represent textual contents. An explanation is provided using Ablation study (which analyzes the importance of each feature).
Data set	ISOT (Guo et al., 2018) and LIAR (Wang, 2017) were the data sets used
Accuracy	99% accuracy for ISOT and 27.4% for LIAR dataset
Critical appraisal of the tool	To detect fake news in social media news items, deep learning techniques and human interpretable components are used. The accuracy of this method has improved over other word embedding-based methods.

Table 10
An overview of the model discussed in Zhou and Zafarani (2019).

Model used	Support vector machine, k-nearest neighbours, naive Bayes, decision tree and random forest. Fake news patterns can help the public comprehend fake news and make fake news detection more understandable.
Data set	For demonstration used two datasets PolitiFact and Buzzfeed.
Accuracy	93% accuracy for politifact and 86% for Buzzfeed data set

may also change. Validation of the model has to be done in news texts in other languages.

3.3. Based on tsetlin machine clauses

The authors of Bhattarai et al. (2021) proposed a unique Tsetlin Machine (TM) based interpretable fake news detection framework. Table 11 presents the details of the model.

Discussion: The present model for detecting fake news is mainly based on textual information. The model's accuracy can be further improved with the incorporation of other auxiliary information from the news material.

3.4. Structured explainability

In the research work (Wu et al., 2021) they considered the broad definition of fake news as "false news", as defined in Zhou and Zafarani

Table 11
An overview of the model based on *Tsetlin machine clauses*.

Model used	The framework takes advantage of the TM's conjunctive clauses to capture the lexical and semantic features of both true and fake news text. Trustworthiness of fake news is measured using clause ensembles. TM clauses contain negated, and unnegated inputs (termed as non-monotone clauses) used for providing explainability.
Evaluation metric for explainability	Explainability of the model is demonstrated using a case study. It provides a higher F1-score than BERT and XLNet, even though it obtained slightly lower accuracy.
Data set	Used Fake News Net data set for demonstration.
Accuracy	Outperforms previous works by at least 5%
Critical appraisal of the tool	The Tsetlin Machine (TM) technique, recently proposed (2018), is used to detect important words and their negations. This method helps to identify deception in text-only news posts.

Table 12
An overview of the research work (Jain and Wallace, 2019).

Model used	They have extracted entity-relation tuples from the training data to build a credential-based multi-relational knowledge graph and then applied a graph convolutional network (GCN) to learn the node and the relation embeddings. Graph convolutional networks classify news into fake or real. The GCN comprises graph embeddings and provides a relational explanation.
Data set	Experimentation done on three datasets Celebrity, PolitiFact and GossipCop.
Accuracy	71.4% , 86.0%, 73.3% accuracy for data sets Celebrity, PolitiFact and GossipCop respectively.
Critical appraisal of the tool	Existing approaches ignore the global semantic aspects of news in favour of concentrating on the features of knowledge graphs. This model goes beyond the contributions of keywords by enhancing the semantic features with structural embeddings from knowledge graphs.

Table 13
Overview of the existing approaches in explainable fake news detection.

Research articles	Features used	Techniques used	Type of explainability
Yang et al. (2019)	News attributes, Semantic meanings of news statements and Linguistic features of news statements	MIMIC, PERT, and ATTN frameworks	Mimic learning (or Explanation by simplification), Perturbation based explanation, Self-attention based explanation and Visual explanation
Shu et al. (2019a) and Cui et al. (2019)	News content and user comments	RNN and Bidirectional GRU	Co-attention captures intrinsic explainability and Visual explanation
Sharma and Sharma (2021)	News content and user comments(filtered)	Multi nominal knowledge base and RNN	Co attention mechanism
Qiao et al. (2020)	Interpretable textual features	CoCoGen, BRNN, and GRU	Ablation explanations
Kurasinski and Mihailescu (2020)	Textual content	BiDir-LSTM-CNN and BERT	Attention based and Visual explanation
Zhou and Zafarani (2019)	Network based patterns	Supervised learning algorithms with five-fold cross-validation	Empirical studies on social psychological theories
Bhattarai et al. (2021)	News contents	Tstelin Machine (TM)	Local and Global interpretability
Wu et al. (2021)	Semantic contents of news articles	Knowledge Graph Framework (KGF)	Structured explainability (Relational Level)
Ni et al. (2021)	Source tweet and retweet users	Graph Attention Network	Attention based explanation
Silva et al. (2021)	User-based, textual and temporal features in propagation pattern	Propagation network	Attention based explanation
Lu and Li (2020)	User-based and textual features	GCAN	Attention based explanation

(2020). Training data contains fake news about public figures and organizations in articles, statements, and speeches. Table 12 presents the details of the model.

Discussion: We find that with the current implementation methodology, early identification of fake news is not possible. The model analyses textual characteristics for fake news detection. If the news spreaders change their writing style, the model will not be able to give accurate results. Recently, fake contents have been created with the inclusion of audio and video. Therefore, analysis of auxiliary features like images, music, and videos will improve the model's accuracy.

4. Observations

We looked at the research on explainable fake news detection methods done between 2018 and 2022. Eleven innovative approaches can be found in this area. Table 13 gives a brief overview of the most recent approaches for detecting explainable fake news, with an emphasis on the characteristics, strategies, and types of explainability that the model makes use of. Fig. 7 illustrates a general analysis of existing solutions for explainable fake news detection.

In this section, we present the findings from our survey. We look at the existing research and discuss it from four different aspects :

(1) Utilizing visual characteristics in explainable fake news detection problems.

(2) Analysis of existing approaches based on explainability types.

(3) Examining current models explainability from the viewpoint of the different explainee.

(4) Taking the evaluation metrics used in current fake news detection models into account.

This categorization and analysis of the existing work in this area is novel and hence is contribution of this review paper. Fig. 8 shows the four perspectives given the literature for explainable fake news detection. Each of the perspectives are detailed in the following Sections.

4.1. Explainable fake news detection using visual features

Most existing efforts in explainable fake news detection explore news content features to provide explanations for fake news detection

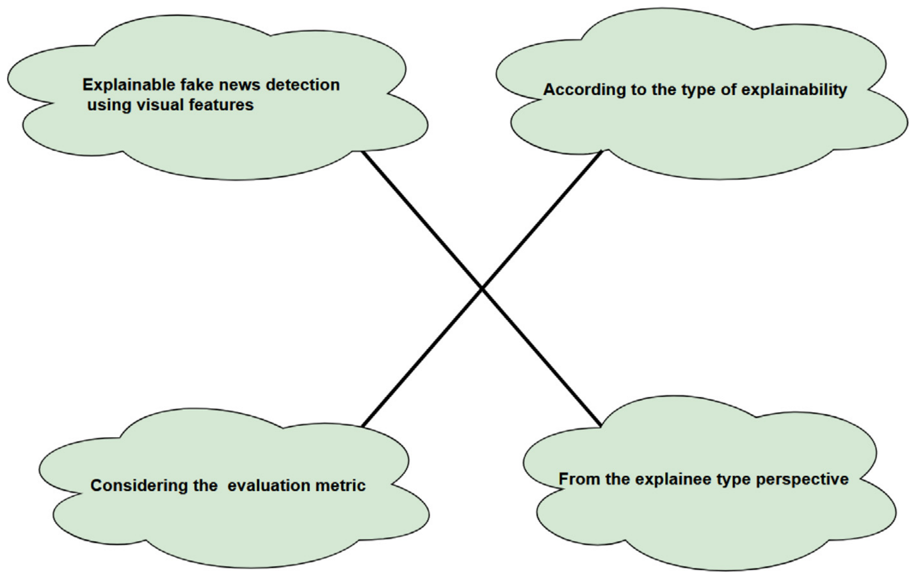

Fig. 8. Perspectives used for analyzing the work on fake news detection with explainability.

problems. Only Shu et al. (2019a), Cui et al. (2019), and Sharma and Sharma (2021) derived explanations for the detection model from the perspective of news content and social engagements of the user. Zhou and Zafarani (2019) takes a different approach, using patterns identified in friendship networks to build an explainable fake news detection system. Recently, Silva et al. (2021) proposed a methodology for explainable fake news identification that leverages user-based, textual, and temporal characteristics in the dissemination pattern of the news.

A majority of recent works, in general, rely on textual analysis. Recently, various approaches Singhal et al. (2019, 2020), and Shah and Kobti (2020) have combined visual-based features with textual features to capture different characteristics of fake news and achieved promising results. Visual content is an essential aspect of fake news since it seeks to use multimedia content with images or videos to entice and mislead readers for quick spread. As far as we know, no studies exist on explainable fake news detection using visual features. Research work needs to be done in this direction.

The authors of Reis et al. (2019) investigated around 300,000 models to identify features' informativeness and suggested that various forms of fake news are identified with specific combinations of features. The authors concluded that various models used significantly different logic to distinguish fake news from true news. Hence combinations of different features may provide better detection and explanations for the predictions. As far as we know, a combination of models with different features for better explainability is unaddressed. As there is no universal definition for "fake news", classifying them under one single definition is not efficient. Each definition covers each aspect of the fake news. Thus based on definition fake news detection problem can be framed as a multi-class problem. So far we have not found any work on a multi-class explainable fake news detection problem that classifies the fake news into distinct classes and offers an explanation based on feature relevance.

4.2. According to the type of explainability

Reviewing from the explainability type perspective, state of the art for explainable fake news detection shows diverse approaches. The most common being intrinsic explainability, and mainly attention mechanisms are used commonly. Yang et al. (2019), Shu et al. (2019a), Cui et al. (2019), Sharma and Sharma (2021), Kurasinski and Mihailescu (2020), Ni et al. (2021), and Silva et al. (2021) provide explanations to the prediction made by the classifier using attention weights. Ni et al. (2021) had used two types of attention networks. Even though they provide accurate explanations, keeps the accuracy uncertain.

Following is a summary of our observations after analyzing the research works listed above.

In the literature, there is a disagreement over whether attention weight should be considered as explainability or not (Serrano and Smith, 2019; Jain and Wallace, 2019). Visual explanation, explanation by simplification (mimic learning), and Feature relevance explanation (ablation explanation, perturbation-based explanation) are some post-hoc explanations types from the literature. Feature relevance explanation only indicates feature's importance, but do not consider information about the feature interactions. These are computationally expensive. Wu et al. (2021) provided a relational explanations for the prediction. While closely looking at the different explanation types, each of them covers different aspects of explainability. Hence combining multiple approaches adds more robustness to the explainability of the model. Only (Yang et al., 2019) tried to provide explainability by combining different approaches. Interactive explanation approaches include interaction between explainee and system, thereby allowing users to improve the system performance. However, in literature, only (Yang et al., 2019) incorporates this approach. Hence we highlight the necessity for more research on interactive explainable detection systems.

4.3. From the explainee type perspective

The type of explainee (End-users, Data experts, or AI experts) and their domain familiarity plays a vital role in deciding the type of explanation to be used. All the existing approaches from the literature prioritize data experts or AI experts. We discovered that these approaches are unfavourable if the explainee are end-users. Thus focus on more human-friendly approaches in explainable fake news detection are essential.

4.4. Considering the evaluation metric

The explainable fake news detection system surveyed employs human evaluation by Amazon Mechanical Turk (AMT) to assess the quality of explanations. Other approaches involve visualization or case study to measure explanation quality. However, we find that the assessment of performance throughout the explainable system is still unresolved. From our observations we have not found any common benchmark on which we can compare the performance of different explainable models.

5. Research directions

Based on our assessment of the related work in this area, we list below some promising research directions that we find in this field:

- Most social media news items employ multimedia content with images or videos to attract and mislead readers for rapid spread. Hence investigating the significance of visual features in identifying explainable fake news is a promising avenue for research.
- Developing explainable models for detecting fake news that combines textual and visual features with the correlation analysis between these features in social media news posts is another potential research domain.
- The fakeness of news articles can be efficiently detected and interpreted by combining various features of news items. Thus an ensemble technique that combines models with different features for better explainability is a promising investigation area.
- There is need to develop a multi-class explainable fake news detection system that classifies the fake news into distinct classes and offers explanations for the predictions made. Such systems can help in detecting fake news regardless of its definition, as there is no universally accepted definition for fake news.

6. Conclusions

In todays world of Online Social Media(OSM), almost everyone is depending on the news items which are published through the OSMs. At the same time, the wide spread of fake news and fake articles have also grown into a very big threat in the domain. Identifying and filtering out fake news is a hot problem which needs urgent research attention. Moreover the Law Enforcement Agencies are looking for convincing and authentic explanations for such classifications of the news items.

In this paper, we have presented a review on existing approaches in the emerging area of explainable AI (XAI) applied to fake news detection and our survey on current methods for explainable fake news detection in the literature. An in depth study on recent approaches in explainable fake news detection is reported. We have categorized the existing work in the area into four and highlighted the research opportunities in all these four groups. In the first category we have highlighted the research opportunities in explainable fake news detection using visual features. In the second category, we discussed opportunities in explainable fake news detection based on the types of explainability. In the third category, we emphasize prospects from the explainee type perspective. In the fourth category, we report the need for research considering the evaluation metric for explainable fake news detection.

Our research focuses on analyzing visual and textual content on social media platforms. Then the relationship between the textual and visual components is used to explain the prediction made.

CRediT authorship contribution statement

Athira A.B.: Conceptualization, Writing – original draft. **S.D. Madhu Kumar:** Supervision. **Anu Mary Chacko:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

References

Atakishiyev, S., Babiker, H., Farruque, N., Goebel1, R., Kima, M.-Y., Motallebi, M.H., Rabelo, J., Syed, T., Zaïane, O.R., 2020. A multi-component framework for the analysis and design of explainable artificial intelligence. arXiv:2005.01908.

Belle, V., Papantonis, I., 2021. Principles and practice of explainable machine learning. Front. Big Data 4, 39. http://dx.doi.org/10.3389/fdata.2021.688969, URL https://www.frontiersin.org/article/10.3389/fdata.2021.688969.

Bhattarai, B., Granmo, O.-C., Jiao, L., 2021. Explainable tsetlin machine framework for fake news detection with credibility score assessment. arXiv:2105.09114.

2021. Claimbuster. https://idir.uta.edu/claimbuster/, Accessed on 20 October.

Classify.news, 2021. FullFact.org. https://makenewscredibleagain.github.io/, Accessed on 20 October.

Cui, L., Shu, K., Wang, S., Lee, D., Liu, H., 2019. DEFEND: A system for explainable fake news detection. In: Proceedings of the 28th ACM International Conference on Information and Knowledge Management. CIKM '19, Association for Computing Machinery, New York, NY, USA, pp. 2961–2964. http://dx.doi.org/10.1145/3357384.3357862.

Du, M., Liu, N., Hu, X., 2019. Techniques for interpretable machine learning. Commun. ACM 63 (1), 68–77. http://dx.doi.org/10.1145/3359786.

2021. Factmata.com. https://factmata.com/, Accessed on 20 October.

2021. FakeNewsNet. https://github.com/KaiDMML/FakeNewsNet, Accessed on 20 October.

2021. Fullfact.org. https://fullfact.org/, Accessed on 20 October.

Guo, H., Cao, J., Zhang, Y., Guo, J., Li, J., 2018. Rumor Detection with Hierarchical Social Attention Network. CIKM '18, Association for Computing Machinery, New York, NY, USA, http://dx.doi.org/10.1145/3269206.3271709.

2021. Hoax-slayer.com. https://smhoaxslayer.com/, Accessed on 20 October.

Jain, S., Wallace, B.C., 2019. Attention is not explanation. CoRR, abs/1902.10186 arXiv:1902.10186 URL http://arxiv.org/abs/1902.10186.

Jin, Z., Cao, J., Zhang, Y., Zhou, J., Tian, Q., 2017. Novel visual and statistical image features for microblogs news verification. IEEE Trans. Multimed. 19 (3), 598–608. http://dx.doi.org/10.1109/TMM.2016.2617078.

Kurasinski, L., Mihailescu, R.-C., 2020. Towards machine learning explainability in text classification for fake news detection. In: 2020 19th IEEE International Conference on Machine Learning and Applications (ICMLA). pp. 775–781. http://dx.doi.org/10.1109/ICMLA51294.2020.00127.

2021. Lead stories. https://leadstories.com/, Accessed on 20 October.

Liu, Y., Wu, Y.-F.B., 2020. FNED: A deep network for fake news early detection on social media. ACM Trans. Inf. Syst. 38 (3), http://dx.doi.org/10.1145/3386253.

Lu, Y.-J., Li, C.-T., 2020. GCAN: Graph-aware co-attention networks for explainable fake news detection on social media. arXiv:2004.11648.

2021. Media bias/fact check. https://mediabiasfactcheck.com/, Accessed on 20 October.

Mohseni, S., Yang, F., Pentyala, S.K., Du, M., Liu, Y., Lupfer, N., Hu, X., Ji, S., Ragan, E.D., 2020. Machine learning explanations to prevent overtrust in fake news detection. CoRR, abs/2007.12358 arXiv:2007.12358 URL https://arxiv.org/abs/2007.12358.

Mohseni, S., Zarei, N., Ragan, E.D., 2021. A multidisciplinary survey and framework for design and evaluation of explainable AI systems. ACM Trans. Interact. Intell. Syst. 11 (3–4), http://dx.doi.org/10.1145/3387166.

Ni, S., Li, J., Kao, H.-Y., 2021. MVAN: Multi-view attention networks for fake news detection on social media. IEEE Access 9, 106907–106917. http://dx.doi.org/10.1109/ACCESS.2021.3100245.

2021. Politifact.com. https://www.politifact.com/, Accessed on 20 October.

Qiao, Y., Wiechmann, D., Kerz, E., 2020. A language-based approach to fake news detection through interpretable features and BRNN. In: Proceedings of the 3rd International Workshop on Rumours and Deception in Social Media (RDSM). Association for Computational Linguistics, Barcelona, Spain (Online), pp. 14–31, URL https://aclanthology.org/2020.rdsm-1.2.

Reis, J.C.S., Correia, A., Murai, F., Veloso, A., Benevenuto, F., 2019. Explainable machine learning for fake news detection. In: Proceedings of the 10th ACM Conference on Web Science. WebSci '19, Association for Computing Machinery, New York, NY, USA, pp. 17–26. http://dx.doi.org/10.1145/3292522.3326027.

Ribeiro, M.T., Singh, S., Guestrin, C., 2016. "Why should I trust you?": Explaining the predictions of any classifier. In: Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. KDD '16, Association for Computing Machinery, New York, NY, USA, pp. 1135–1144. http://dx.doi.org/10.1145/2939672.2939778.

Ruchansky, N., Seo, S., Liu, Y., 2017. CSI: A hybrid deep model for fake news detection. In: Proceedings of the 2017 ACM on Conference on Information and Knowledge Management. CIKM '17, Association for Computing Machinery, New York, NY, USA, pp. 797–806. http://dx.doi.org/10.1145/3132847.3132877.

Selvaraju, R.R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., Batra, D., 2017. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In: 2017 IEEE International Conference on Computer Vision (ICCV). pp. 618–626. http://dx.doi.org/10.1109/ICCV.2017.74.

Serrano, S., Smith, N.A., 2019. Is attention interpretable?. arXiv:1906.03731.

Shah, P., Kobti, Z., 2020. Multimodal fake news detection using a cultural algorithm with situational and normative knowledge. In: 2020 IEEE Congress on Evolutionary Computation (CEC). pp. 1–7. http://dx.doi.org/10.1109/CEC48606.2020.9185643.

Sharma, D.K., Sharma, S., 2021. Comment filtering based explainable fake news detection. In: Singh, P.K., Wierzchoń, S.T., Tanwar, S., Ganzha, M., Rodrigues, J.J.P.C. (Eds.), Proceedings of Second International Conference on Computing, Communications, and Cyber-Security. Springer Singapore, Singapore, pp. 447–458.

Shu, K., Cui, L., Wang, S., Lee, D., Liu, H., 2019a. DEFEND: Explainable fake news detection. In: Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery &Amp; Data Mining. KDD '19, Association for Computing Machinery, New York, NY, USA, pp. 395–405. http://dx.doi.org/10.1145/3292500.3330935.

Shu, K., Mahudeswaran, D., Wang, S., Liu, H., 2019b. Hierarchical propagation networks for fake news detection: Investigation and exploitation. CoRR, abs/1903.09196 arXiv:1903.09196 URL http://arxiv.org/abs/1903.09196.

Shu, K., Wang, S., Liu, H., 2018. Understanding user profiles on social media for fake news detection. In: 2018 IEEE Conference on Multimedia Information Processing and Retrieval (MIPR). pp. 430–435. http://dx.doi.org/10.1109/MIPR.2018.00092.

Shu, K., Zhou, X., Wang, S., Zafarani, R., Liu, H., 2019c. The Role of User Profiles for Fake News Detection. ASONAM '19, Association for Computing Machinery, New York, NY, USA, pp. 436–439. http://dx.doi.org/10.1145/3341161.3342927.

Silva, A., Han, Y., Luo, L., Karunasekera, S., Leckie, C., 2021. Propagation2Vec: Embedding partial propagation networks for explainable fake news early detection. Inf. Process. Manage. 58 (5), 102618. http://dx.doi.org/10.1016/j.ipm.2021.102618, URL https://www.sciencedirect.com/science/article/pii/S030645732100114X.

Singh, V.K., Ghosh, I., Sonagara, D., 2021. Detecting fake news stories via multimodal analysis. J. Assoc. Inf. Sci. Technol. 72 (1), 3–17. http://dx.doi.org/10.1002/asi.24359, arXiv:https://asistdl.onlinelibrary.wiley.com/doi/pdf/10.1002/asi.24359 URL https://asistdl.onlinelibrary.wiley.com/doi/abs/10.1002/asi.24359.

Singhal, S., Kabra, A., Sharma, M., Shah, R.R., Chakraborty, T., Kumaraguru, P., 2020. SpotFake+: A multimodal framework for fake news detection via transfer learning (student abstract). In: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 34 (10). pp. 13915–13916. http://dx.doi.org/10.1609/aaai.v34i10.7230, URL https://ojs.aaai.org/index.php/AAAI/article/view/7230.

Singhal, S., Shah, R.R., Chakraborty, T., Kumaraguru, P., Satoh, S., 2019. SpotFake: A multi-modal framework for fake news detection. In: 2019 IEEE Fifth International Conference on Multimedia Big Data (BigMM). pp. 39–47. http://dx.doi.org/10.1109/BigMM.2019.00-44.

2021. Snopes.com. https://www.snopes.com/, Accessed on 20 October.

Tanwar, V., Sharma, K., 2020. Multi-model fake news detection based on concatenation of visual latent features. In: 2020 International Conference on Communication and Signal Processing (ICCSP). pp. 1344–1348. http://dx.doi.org/10.1109/ICCSP48568.2020.9182398.

2021. Truthorfiction.com. https://www.truthorfiction.com/, Accessed on 20 October.

Wang, W.Y., 2017. "Liar, liar pants on fire": A new benchmark dataset for fake news detection. arXiv:1705.00648.

Wikipedia, 2021. Fake news. URL https://en.wikipedia.org/wiki/Fake_news#Definition, Accessed on 20 October 2021.

Wu, K., Yuan, X., Ning, Y., 2021. Incorporating relational knowledge in explainable fake news detection. In: Advances in Knowledge Discovery and Data Mining. Springer International Publishing, Cham, pp. 403–415.

2021. Xgboost. https://github.com/dmlc/xgboost, Accessed on 20 October.

Yang, F., Pentyala, S.K., Mohseni, S., Du, M., Yuan, H., Linder, R., Ragan, E.D., Ji, S., Hu, X.B., 2019. XFake: Explainable fake news detector with visualizations. In: The World Wide Web Conference. WWW '19, Association for Computing Machinery, New York, NY, USA, pp. 3600–3604. http://dx.doi.org/10.1145/3308558.3314119.

Zhang, X., Ghorbani, A.A., 2020. An overview of online fake news: Characterization, detection, and discussion. Inf. Process. Manage. 57 (2), 102025. http://dx.doi.org/10.1016/j.ipm.2019.03.004, URL https://www.sciencedirect.com/science/article/pii/S0306457318306794.

Zhou, X., Zafarani, R., 2019. Network-based fake news detection: A pattern-driven approach. SIGKDD Explor. Newsl. 21 (2), 48–60. http://dx.doi.org/10.1145/3373464.3373473.

Zhou, X., Zafarani, R., 2020. A survey of fake news: Fundamental theories, detection methods, and opportunities. ACM Comput. Surv. 53 (5), http://dx.doi.org/10.1145/3395046.